

Lectures on Multiphysics Modeling

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Fluids 1

24-30-31

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Topics

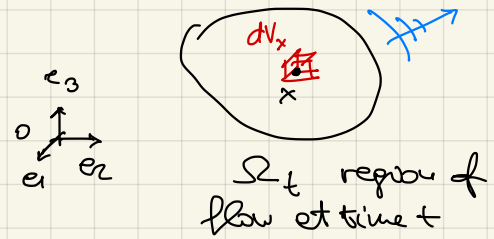
- Fluid Dynamics
(Navier-Stokes, Stokes)
- Linear Elasticity
- Nonlinear Elasticity
- Fluid-Structure Interaction

Schedule

Monday's	8.30 - 11.30	A 26
Tuesday's	16.30 - 18.30	A 28

Lecture 1 Navier-Stokes equations (for incompressible fluids)

A nonlinear system of 4 scalar PDE's



$$\begin{cases} \rho \left[\frac{\partial}{\partial t} \underline{v} + (\underline{v} \cdot \nabla) \underline{v} \right] = \mu \Delta \underline{v} - \nabla p + \rho \underline{g} \\ \operatorname{div} \underline{v} = 0 \end{cases}$$

$x = (x_1, x_2, x_3)$ position in space

$\underline{v}(x, t) = (v_1, v_2, v_3)$ velocity of fluid particle occupying place x at time t
 $p(x, t)$ pressure at x, t

4 scalar unknown functions (x, t)

$\rho > 0$ density, $\mu > 0$ viscosity, two constants (constant ρ from {homogeneity, incompressibility})
 $\underline{g}$ acceleration of gravity ($= -g \underline{e}_3$, g constant)
 These are data

Context: classical Newtonian mechanics for a class of fluids which describes well water and air in non-extreme conditions.

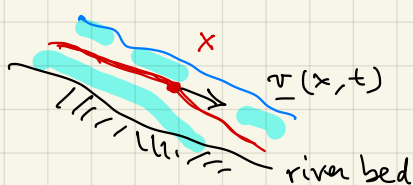
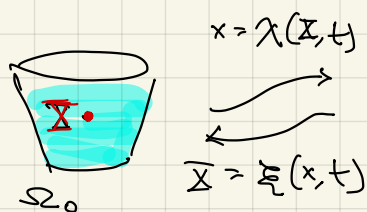
$$\begin{cases} \rho \underline{a} = \operatorname{div} \underline{T} + \rho \underline{g} \quad \rightarrow \quad \underline{a} \overbrace{\rho dV_x}^{dV_x} = (\operatorname{div} \underline{T}) dV_x + \underline{g} \overbrace{\rho dV_x}^{dV_x} \\ 0 = \operatorname{div} \underline{v} \quad \rightarrow \quad \text{incompressibility} \end{cases}$$

$\rightarrow$ " $m \underline{a} = \underline{f}$ " for a fluid particle

$\underline{a}(x, t)$ acceleration of fluid particle occupying place x at time t

In a sense to be made precise, it's $m \underline{a} = \underline{f}$ for a fluid particle (elementary volume dV_x)

Let's start with the acceleration a



Current configuration $\Omega_t = \chi(\Omega_0, t)$
(at time t)

$[\Omega_0, \Omega_t]$

Look at X : (Lagrangian description) Particles are tagged by their position in the reference config. What happens to particle X ?

Look at x : (Eulerian description) Focus on x , a location in space. What happens at place x ?

$x = \chi(X, t)$ place occupied by X at time t

$X = \xi(x, t)$ particle that occupies place x at time t

χ and ξ are inverse of each other: $\chi(\xi(x, t), t) = x$, $\xi(\chi(X, t), t) = X$

(read these in words)
[or $f(f^{-1}(x)) = x$]

Look at X : $\chi(X, t)$ motion of X

$\dot{\chi}(X, t)$ velocity of X at time t

$\ddot{\chi}(X, t)$ acceleration of X at time t

$\left[\dot{\chi}(X, t) = \frac{\partial}{\partial t} \chi(X, t) \right.$
at fixed particle]

Look at x : $\underline{v}(x, t)$ velocity of the particle that occupies place x at time t

$\underline{a}(x, t)$ acceleration " " " " " " " "

Therefore $\underline{v}(x, t) = \dot{\chi}(X, t) \Big|_{X = \xi(x, t)}$

$\underline{a}(x, t) = \ddot{\chi}(X, t) \Big|_{X = \xi(x, t)}$

The big result is that we can compute a from v !

$$\underline{v}(x, t) = \dot{\underline{\chi}}(\underline{x}, t) \Big|_{\underline{x} = \underline{\xi}(x, t)}$$

$$\underline{a}(x, t) = \ddot{\underline{\chi}}(\underline{x}, t) \Big|_{\underline{x} = \underline{\xi}(x, t)}$$

Set $x = \chi(\underline{x}, t)$ and use $\underline{\xi}(\chi(\underline{x}, t), t) = \underline{x}$, to get

$$\underline{v}(\chi(\underline{x}, t), t) = \dot{\underline{\chi}}(\underline{x}, t) \quad (*)$$

$$\underline{a}(x, t) = \ddot{\underline{\chi}}(\underline{x}, t) \Big|_{\underline{x} = \underline{\xi}(x, t)} = \frac{d}{dt} \left(\underline{v}(\chi(\underline{x}, t), t) \right) \Big|_{\underline{x} = \underline{\xi}(x, t)}$$

$\underline{a}$ is also denoted $\dot{\underline{v}}$ (material time derivative, i.e. time derivative of fixed particle)

$$\dot{\underline{v}}(x, t) = \frac{d}{dt} \underline{v}(\chi(\underline{x}, t), t) \Big|_{\underline{x} = \underline{\xi}(x, t)}$$

Chain rule

$$x = (x_1, x_2, x_3)$$

$$= \chi(\underline{x}, t) = (\chi_1(\underline{x}, t), \chi_2(\underline{x}, t), \chi_3(\underline{x}, t))$$

$$\underline{v}(x, t) = \underline{v}(x_1, x_2, x_3, t) \rightarrow \underline{v}(\chi_1(\underline{x}, t), \chi_2(\underline{x}, t), \chi_3(\underline{x}, t), t)$$

$$\dot{\underline{v}}(x, t) = \left(\frac{\partial}{\partial t} \underline{v}(\underline{\chi}(\underline{x}, t), t) + \sum_{j=1}^3 \frac{\partial}{\partial x_j} \underline{v}(\underline{\chi}(\underline{x}, t), t) \dot{\underline{\chi}}_j(\underline{x}, t) \right) \Big|_{\underline{x} = \underline{\xi}(x, t)}$$

$$= \frac{\partial}{\partial t} \underline{v}(x, t) + \sum_{j=1}^3 \frac{\partial}{\partial x_j} \underline{v}(x, t) \underline{v}_j(x, t)$$

$$= \frac{\partial}{\partial t} \underline{v} + (\underline{v} \cdot \nabla_x) \underline{v}$$

[algorithmic]

$$= \frac{\partial}{\partial t} \underline{v} + (\nabla_x \underline{v}) \underline{v}$$

[physical]

a vector; its i -th component is $\sum_{j=1}^3 \frac{\partial}{\partial x_j} v_i v_j = \sum_{j=1}^3 (\nabla_x \underline{v})_{ij} v_j$

It's a product row times column of a matrix $[\nabla_x \underline{v}]$ and a column vector $\underline{v}$.

$$\left[((\nabla_x \underline{v}) \underline{v})_i \right]_{3 \times 1} = \left[\frac{\partial v_i}{\partial x_j} \right]_{3 \times 3} \left[v_j \right]_{3 \times 1} = \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_1}{\partial x_2} & \frac{\partial v_1}{\partial x_3} \\ \frac{\partial v_2}{\partial x_1} & \frac{\partial v_2}{\partial x_2} & \frac{\partial v_2}{\partial x_3} \\ \frac{\partial v_3}{\partial x_1} & \frac{\partial v_3}{\partial x_2} & \frac{\partial v_3}{\partial x_3} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

← this is really $i=2$

In other words the gradient of the velocity vector field is a matrix with components $(\nabla_x \underline{v})_{ij} = \frac{\partial v_i}{\partial x_j}$

$$\delta \underline{v} = \dot{\underline{v}} dt = \frac{\partial}{\partial t} \underline{v} dt + \underbrace{(\nabla_x \underline{v}) \underline{v} dt}_{\text{displacement during } dt}$$

variation of $\underline{v}$ when particle moves from x to $x + \underline{v} dt$

Taylor expansion:

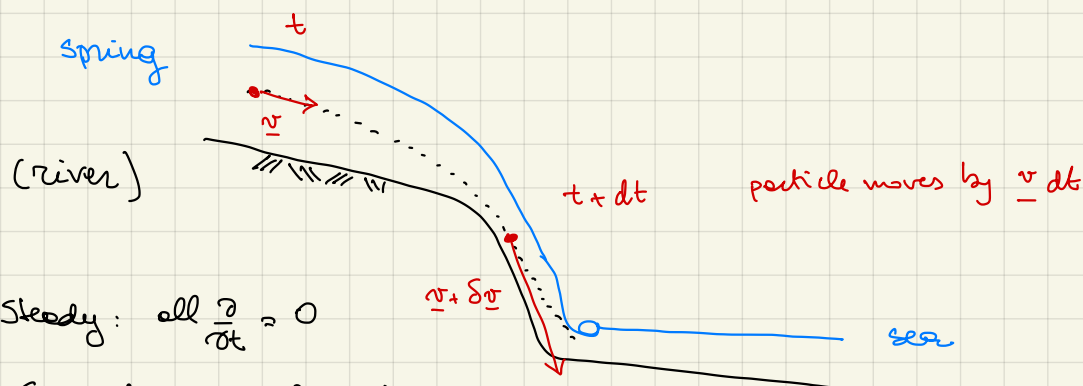
$$1D \quad f(x+h) = f(x) + \left(\frac{d}{dx} f(x) \right) h + o(h)$$

$$3D \quad \underline{u}(x+\underline{h}) = \underline{u}(x) + \left[\nabla_x \underline{u}(x) \right] \left[\underline{h} \right] + o(h)$$

for velocity: $\underline{v}(x+\underline{h}) - \underline{v}(x) = \left[\nabla_x \underline{v}(x) \right] \left[\underline{h} \right] + o(h)$ when $\underline{h} = \underline{v}(x) dt$

Finally $\dot{\underline{v}} = \underbrace{\frac{\partial}{\partial t} \underline{v}}_{\text{local acceleration } \delta \underline{v}_{\text{local}}} + \underbrace{(\nabla_x \underline{v}) \underline{v}}_{\text{convective acceleration } \delta \underline{v}_{\text{conv}}}$ can be computed only from $\underline{v}$ (without knowledge of x !!)

$$\dot{\underline{v}} dt = \delta \underline{v} = \frac{\partial \underline{v}}{\partial t} dt + (\nabla_x \underline{v}) \underline{v} dt \rightarrow \text{Interpretation}$$



Steady: all $\frac{\partial}{\partial t} = 0$

Convective: move from spring to sea
Experience change of velocity

Local: associated with unsteadiness. Consider velocities summer/winter

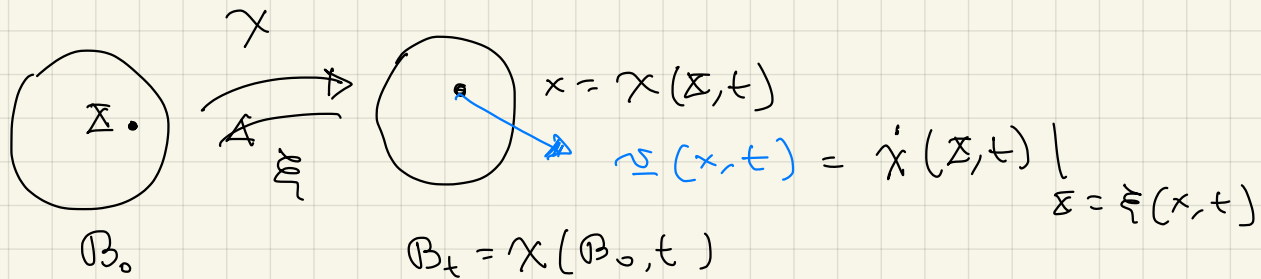
[Newton, Leibnitz, Euler]

In summary (material vs spatial time derivative)

$$\dot{\underline{v}} = \frac{\partial}{\partial t} \underline{v} + (\nabla_x \underline{v}) \underline{v}$$

Other important kinematic relations (material vs spatial gradient)

$$\nabla_{\underline{x}} = \frac{\partial}{\partial \underline{x}} \quad \text{vs} \quad \nabla_x = \frac{\partial}{\partial x}$$



Notation

$$\underline{F} = \nabla_{\underline{x}} \chi \quad (\text{deformation gradient})$$

$$J = \det \underline{F} \quad (\text{Jacobian determinant})$$

Results

$$\dot{\underline{F}} \underline{F}^{-1} = \nabla_x \underline{v} \bigg|_{\underline{x} = \chi(\underline{X}, t)}$$

$$\dot{\underline{F}} = \nabla_x \underline{v} \bigg|_{\underline{x} = \chi(\underline{X}, t)} \underline{F}$$

$$\frac{\partial}{\partial \underline{x}} \dot{\chi} = \frac{\partial}{\partial x} \dot{\chi} \frac{\partial x}{\partial \underline{x}}$$

$$\begin{aligned} \dot{J} &= J \operatorname{tr}(\dot{\underline{F}} \underline{F}^{-1}) \\ &= J \operatorname{div} \underline{v} \bigg|_{\underline{x} = \chi(\underline{X}, t)} \end{aligned}$$

useful for incompressibility in fluids

$$\begin{aligned} \dot{J} &= J \operatorname{tr}(\dot{\underline{F}} \underline{F}^{-1}) = J \dot{\underline{F}} \underline{F}^{-1} \cdot \underline{I} = J \dot{\underline{F}} \cdot \underline{F}^{-T} \\ &= J \underline{F}^{-T} \cdot \dot{\underline{F}} \end{aligned}$$

useful for incompressibility in solids
(non linear elasticity)

Einstein convention: sum over repeated indices tacitly understood

where $\underline{A} \cdot \underline{B} = \sum_{ij} A_{ij} B_{ij}$, $\underline{A} \cdot \underline{B} \cdot \underline{C} = \underline{A} \cdot \underline{B} \underline{C}^T$ from $(A_{ik} B_{kj}) C_{ij} = A_{ik} \underbrace{(C_{ij} (\underline{B}^T)_{jk})}_{(\underline{CB}^T)_{ik}}$